Debounce Test Sheet / Test Log

Webinar 2 — GPIO / Interrupt / Button Bouncing

Environment

Параметр	Значення
Board	ESP32 DevKit v1
Input	BUTTON / NOISE → GPIO18
Output	LED → GPIO2
Simulator	Wokwi
Serial	115200
Build	BUGGY

Acceptance Criteria

Valid physical presses = logical press events. RAW interrupts may be greater. Short unstable noise must not create a false logical press/action.

Test Cases

ID	Name	Precondition	Action / Steps	Expected Result	Actual Result	P/F	Bug ID
TC-01	Normal press	Wokwi Simulator launched	Press BUTTON	1 press → 1 logical event → 1 action	1 press → 1 logical event → 1 action	P	
TC-02	Bounce	Wokwi Simulator launched	Press BUTTON	D0 may contain multiple transitions; D1 changes only once for accepted press. 1 logical event and 1 action	A single press produced 44 RAW edges (interrupts) on D0 due to contact bounce, but D1 (LOGICAL) changed only once (a single debounced event). This resulted in exactly 1 logical event and 1 action, as expected.	P	
TC-03	Long press	Wokwi Simulator launched	Hold BUTTON	No repeated press events/actions while held	No repeated press events/actions while held; 1 action and 1 logical event were created	P	

TC-04	5 rapid presses	Wokwi Simulator launched	Press BUTTON 5 times	5 logical press events → 5 actions	5 logical press events → 5 actions	P	
TC-05	Noise pulse	Wokwi Simulator launched	Press NOISE	No false logical event/action	1 logical event/action	F	BUG-001
TC-06	Noise during critical operation	Wokwi Simulator launched	1. Press BUTTON 2. Press NOISE	No extra logical event/action; operation continues	New logical event and action; Critical operation finished	F	BUG-002

Execution Log

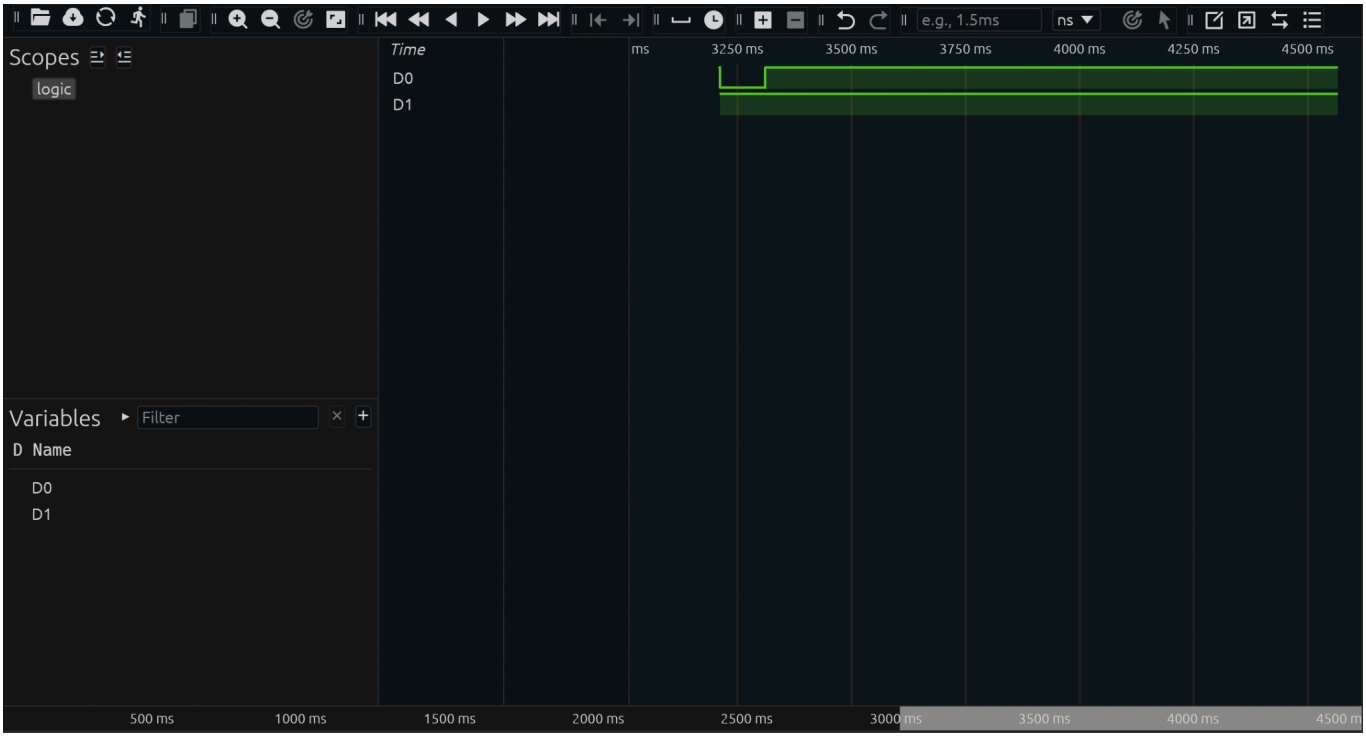
ID	Scenario	Physical presses	RAW	Logical	Actions	Expected	Actual	P/F
TC-01	Normal press	1	47	1	1	1	1	Pass
TC-02	Natural/Short bounce	1	44	1	1	1	1	Pass
TC-03	Heavy bounce							Not executed / Covered by TC-02
TC-04	Long press	1	35	1	1	1	1	Pass
TC-05	Rapid presses	5	660	5	5	5	5	Pass
TC-06	Short noise pulse	0	1	1	1	0	1	Fail
TC-07	Noise during critical operation	1	158	2	2	1	2	Fail

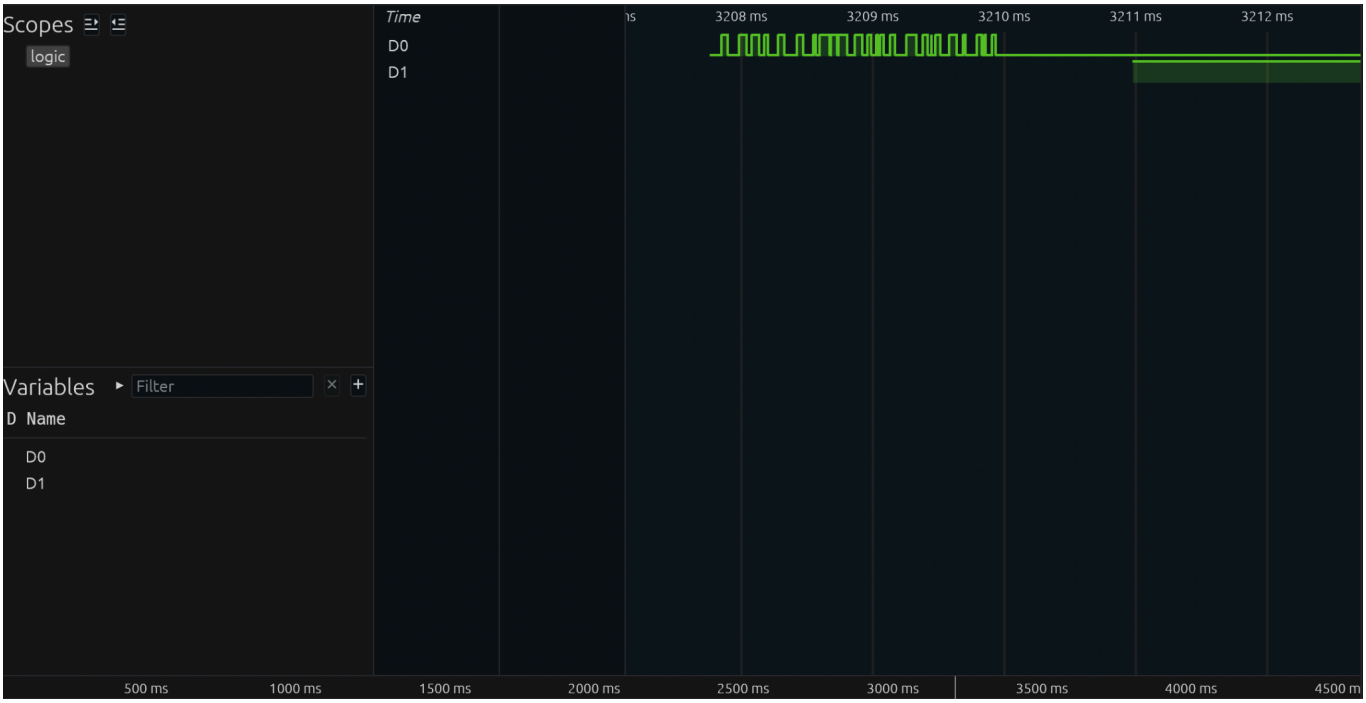
Comments on Execution Log TC-03 (Heavy bounce):

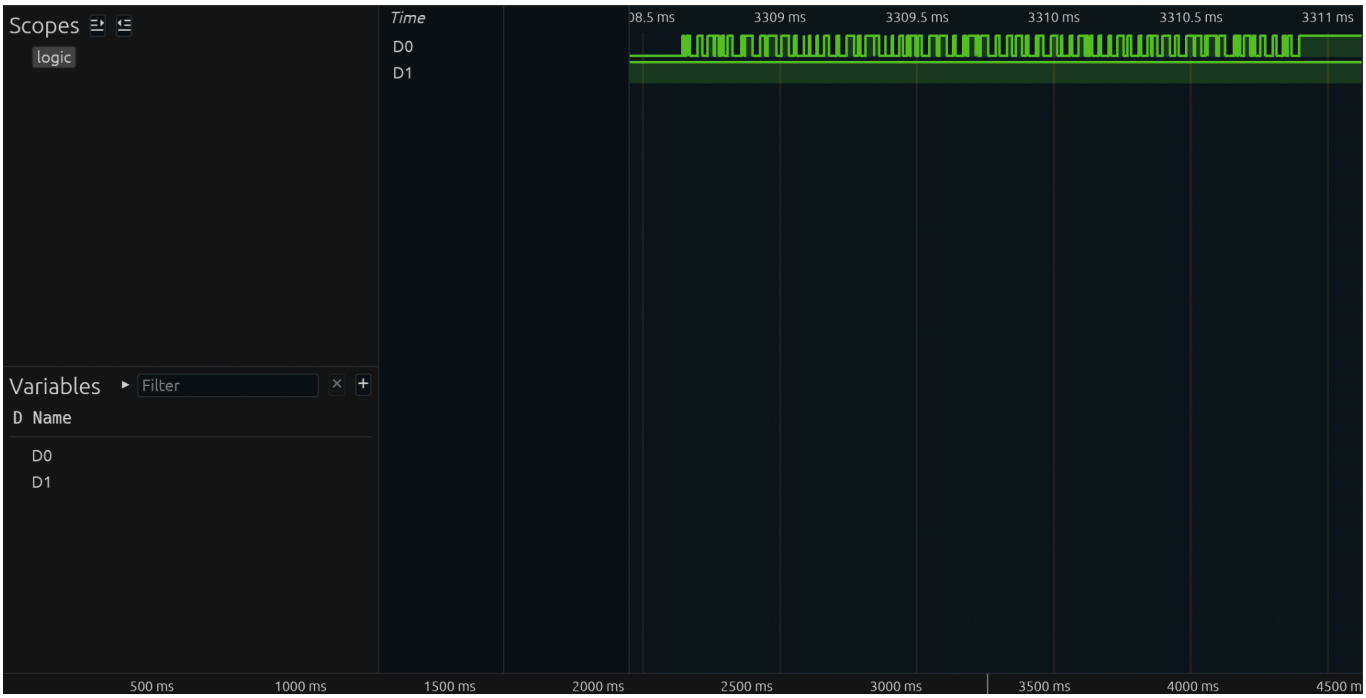
TC-03 not executed. The test bench does not allow controlling the intensity of contact bounce, so a heave bounce could not be reproduced reliably. This scenario is covered by TC-02 (natural/short bounce).

Button bounce: D0 RAW vs D1 LOGICAL (Logic Analyzer)

The waveform shows that a single physical press causes contact bounce on D0 (RAW): 44 edges (interrupts) were registered. A similar bounce is also visible on button release. In contrast, D1 (LOGICAL) shows only one clean transition, i.e. one debounced event, which produced exactly 1 logical event and 1 action. This confirms that the debounce logic works correctly.

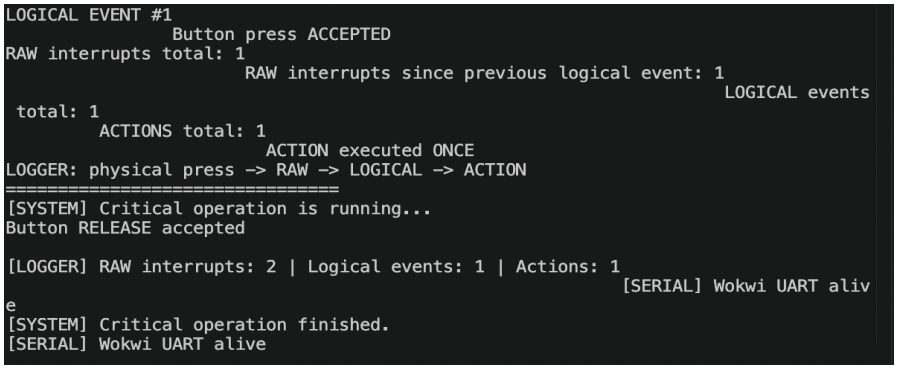


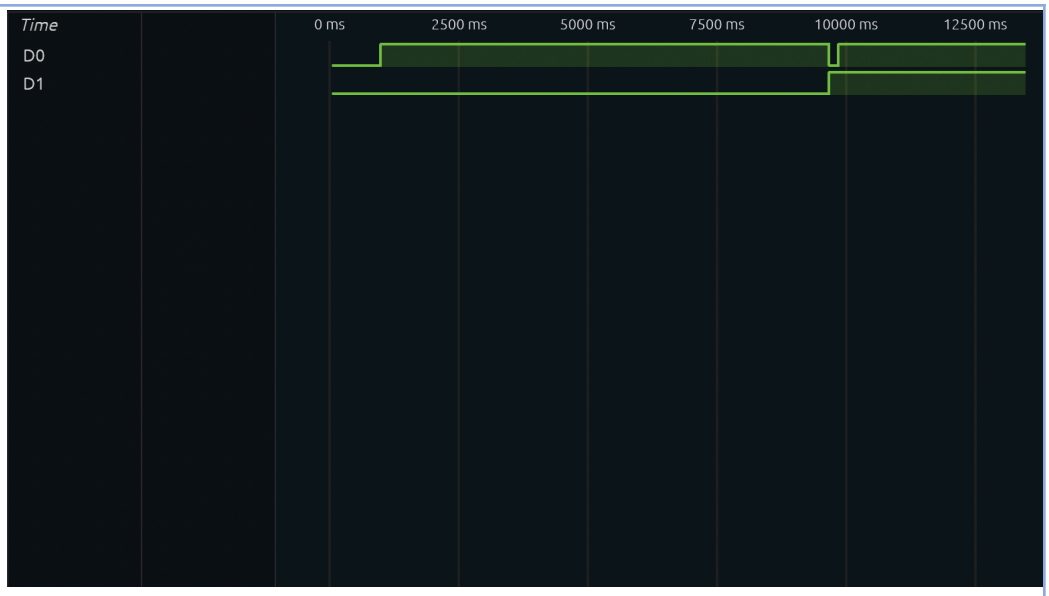




Bug Reports

Bug ID	BUG-001
Title	1 ms debounce window is too short to filter out actual noise

Precondition	Open the project in VS Code and launch the Wokwi simulator
Steps	1. Generate a short noise pulse by clicking the NOISE button
Expected	No logical events and actions
Actual	The pulse is far longer than the debounce window (1 ms, DEBOUNCE_US = 1000), so the debounce logic treated it as a valid press.
Severity	Major
Reproducibility	5/5 Reproducible
Environment	MacBook Pro, macOS Tahoe 26.5.1, VS Code 1.138.0, Wokwi Simulator, ESP32 DevKit v1
Evidence / Notes	Measurement: NOISE pulse duration on D0 ≈ 182.68 ms (Δt between falling and rising edge, measured with cursors in Surfer). D1 (LOGICAL) switched once. Analysis: The pulse is far longer than the debounce window (1 ms, DEBOUNCE_US = 1000), so the debounce logic treated it as a valid press. Limitation: The NOISE input was triggered manually with a mouse click, which typically produces pulses of tens to hundreds of ms. A truly short noise spike (shorter than the debounce window) cannot be reproduced this way, so this manual test cannot verify rejection of short noise. 

	 Video: https://drive.google.com/file/d/1C-e4oFAvUVFmbaIOOZHqCEOvrrjQEkvT/view?usp=sharing

Bug ID	BUG-002
Title	Noise during critical operation creates a new logical event/action and finishes a critical operation
Precondition	Open the project in VS Code and launch the Wokwi simulator
Steps	1. Press BUTTON 2. Press NOISE
Expected	No logical events and actions; Critical operation continues
Actual	New logical event and action; Critical operation finished
Severity	Major
Reproducibility	5/5 Reproducible
Environment	MacBook Pro, macOS Tahoe 26.5.1, VS Code 1.138.0, Wokwi Simulator, ESP32 DevKit v1

Evidence / Notes	

```
LOGICAL EVENT #1
               Button press ACCEPTED
RAW interrupts total: 63
                      RAW interrupts since previous logical event: 63
                                                                    LOGICAL even
ts total: 1
          ACTIONS total: 1
                           ACTION executed ONCE
LOGGER: physical press -> RAW -> LOGICAL -> ACTION
================================
[SYSTEM] Critical operation is running...
Button RELEASE accepted

[LOGGER] RAW interrupts: 157 | Logical events: 1 | Actions: 1

================================
LOGICAL EVENT #2
               Button press ACCEPTED
RAW interrupts total: 158
                       RAW interrupts since previous logical event: 95
                                                                     LOGICAL eve
nts total: 2
           ACTIONS total: 2
                            ACTION executed ONCE
LOGGER: physical press -> RAW -> LOGICAL -> ACTION
================================
Button RELEASE accepted
[SERIAL] Wokwi UART alive
[SYSTEM] Critical operation finished.

[LOGGER] RAW interrupts: 159 | Logical events: 2 | Actions: 2
```

Video: https://drive.google.com/file/d/1h1pDw7O14OyGo1JCm6XU98w8dqnANzkA/view?usp=sharing

Requirements clarification is required:

- Specify the expected LED behavior when a new logical event is added.

QA Conclusion: Fail

Debounce logic was tested with normal press, bounce, and noise scenarios. Debounce correctly filters button bounce (many RAW transitions result in one logical event and one action), but the system is not ready for release because a short noise pulse creates an extra logical event and action and prematurely terminates a critical operation that should continue according to the requirements.

The measured NOISE pulse lasted approximately 182.68 ms, significantly exceeding the 1 ms debounce window; therefore, this manual test could not verify rejection of genuinely short noise spikes.

These issues must be investigated and fixed.